



Graph Algorithms and Applications

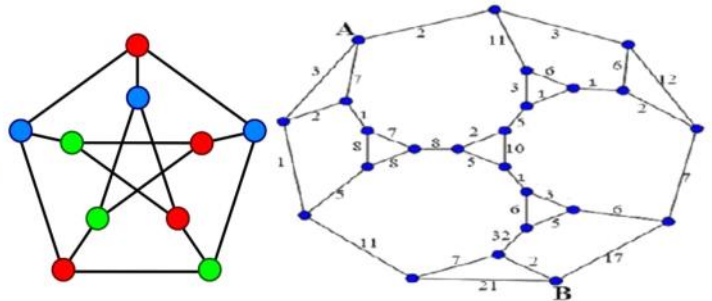
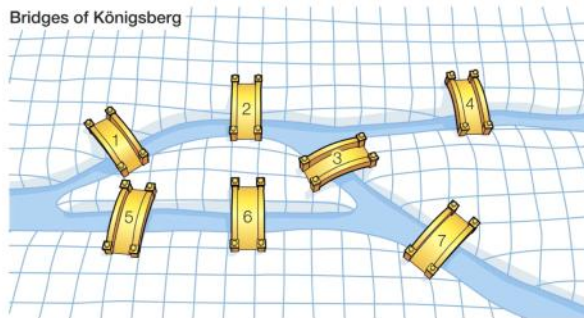
Course Code: 18B14CI645

Credit: 3

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Graph Theory: The Language of Connections

Edge Coloring

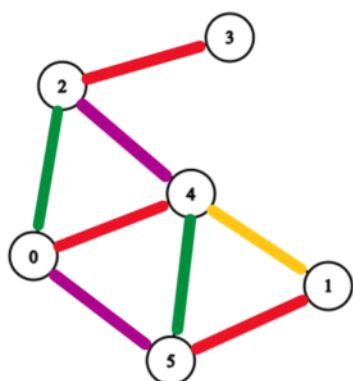
Terminology for edge coloring is similar to vertex coloring.
i.e.

Edge k -coloring

Assigning k -colors to the edges of a graph such
that :- No 2 adjacent edges have same color.



Proper edge coloring



Proper edge coloring

Edge color class

Subset of the set of the edges of a particular graph such that all the edges in that subset have the same color.

red $\{23, 04, 51\}$

green $\{20, 45\}$

purp. $\{24, 05\}$

yel. $\{41\}$

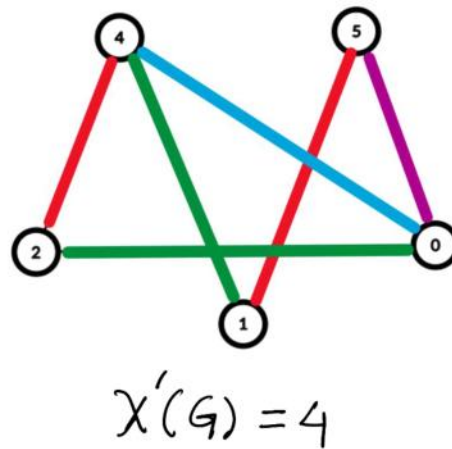
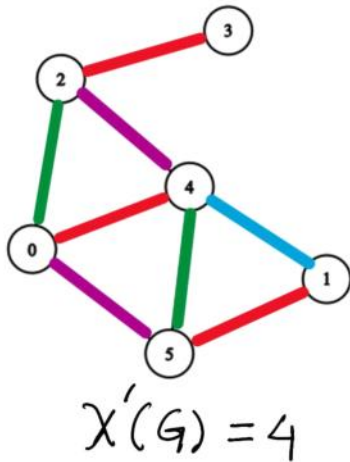
parallel edges

Not a fⁿ of vertex coloring

Strong fⁿ of Edge coloring.

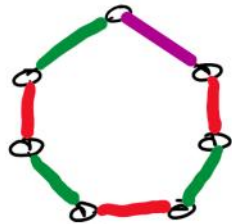
Edge – Chromatic Number $\chi'(G)$

It is the min. no. of different colors required for proper edge coloring.





$$\chi'(G) = 3$$

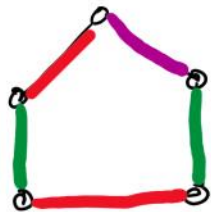


$$\chi'(G) = 3$$

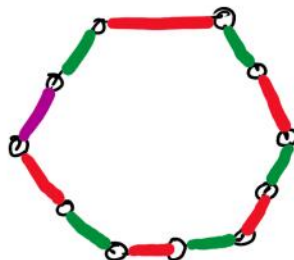
Name?

Odd cycle

Evaluate $\chi'(G)$ for others

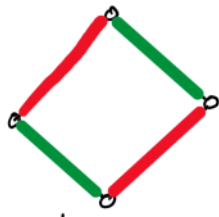


$$\chi'(G) = 3$$

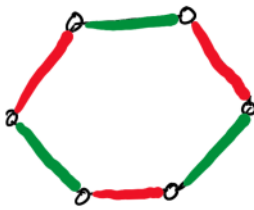


$$\chi'(G) = 3$$

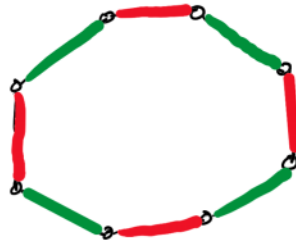
$$\chi'(C_{2n+1}) = 3$$



$$\chi'(G) = 2$$



$$\chi'(G) = 2$$



$$\chi'(G) = 2$$

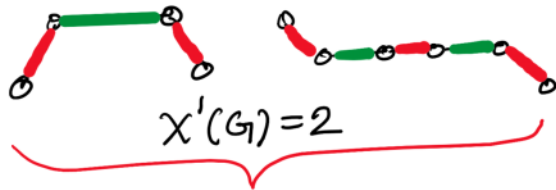
Name?

Even cycle

Evaluate $\chi'(G)$ for others

$$\chi'(C_{2n}) = 2$$

Evaluate $\chi'(G)$ for :-



Path graph
($n \geq 3$)

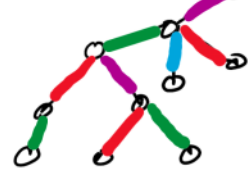
$$\chi'(P_n) = 2$$

↑
 $n \geq 3$

$$\chi'(G) = 3$$



$$\chi'(G) = 4$$



Tree

What is the max. deg. in graphs?

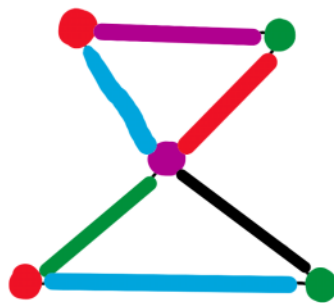
$$\chi'(T) = \Delta(T)$$

↑
max. deg in T

Graph Coloring

Combining both vertex and Edge coloring.

A **proper total coloring** of a graph is an assignment of a color to each vertex and each edge so that **no pair of adjacent vertices or adjacent edges receive the same color** and **every edge is colored differently from its end points**.

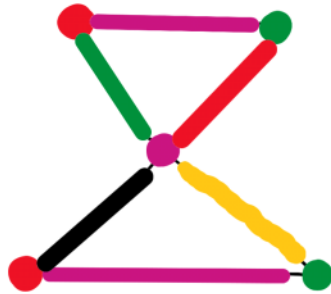


Proper vertex col.

Proper edge col. + Graph.

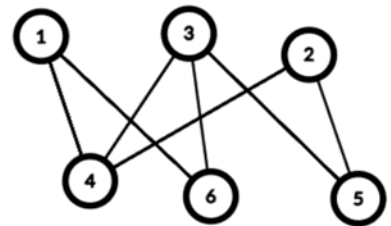
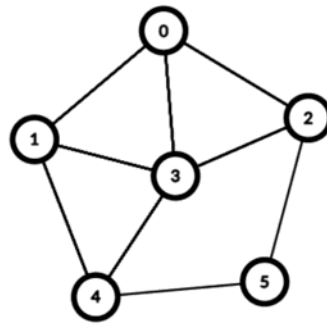
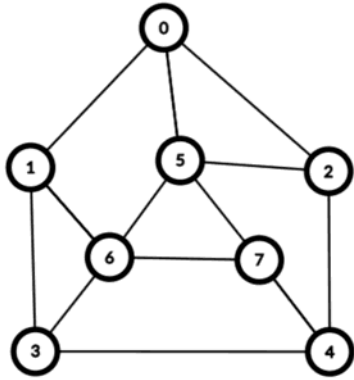
Graph – Chromatic Number

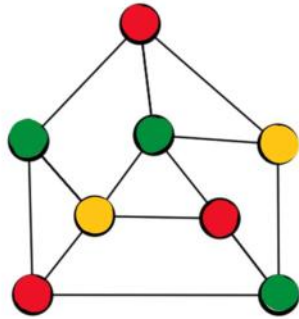
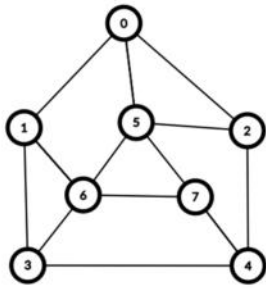
Least no. of colors required for a proper coloring of a graph.



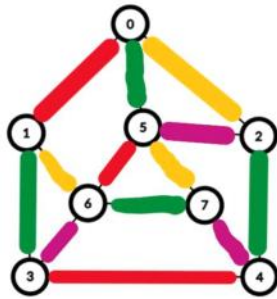
$$\chi(G) = 5$$

For the following graphs do the edge, vertex and graph coloring. Also, find $\chi(G)$, $\chi'(G)$, and $\chi''(G)$.

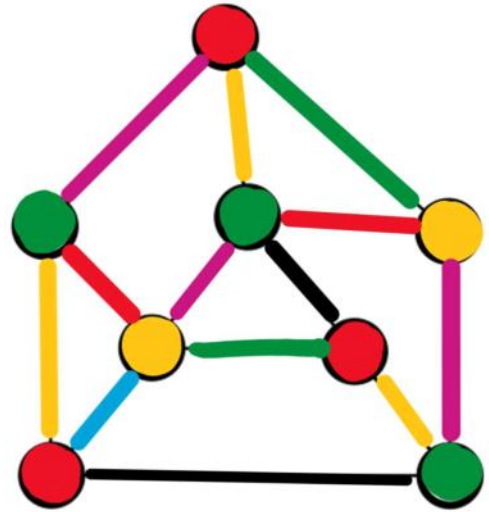




$$\chi(G) = 3$$



$$\chi'(G) = 4$$



$$\chi''(G) = 6$$